# The first internal orbit at odd-degree equality

Human Compression project: L044 continuation

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le \mathrm{Alt}(\Omega)$ be transitive with a connected string C-marking and all maximal omissions intransitive. Assume $n = 2r + 1$ and the actual transitive six-point head on $D = \{1, \dots, 6\}$, with the literal interface

$$\tau = \rho_3 = (6\ b)\delta, \quad R = G_{\ge 3} = \mathrm{Alt}(\Omega \setminus \{1, \dots, 5\}), \quad \sigma = \rho_4, \quad \sigma|_D = 1.$$

The involution $\delta$ fixes $D, b$. Put $H = G_{\ge 5}$ and $K = G_{>5}$, and suppose $H$ fixes exactly $D \cup \{b\}$. Use actual normalized assignments of two full ambient-eligible edges per tail colour to a single $H$-orbit, as in the acquired assignment theorem.

**Theorem 1.** *Suppose the two nontrivial $H$-orbits are $O$ of size $2q$ and $V$ of size $2s$, where $q \ge 3$ is odd and $s \ge 1$. Assume $\delta$ preserves $O$ and acts freely there, and that their assigned sets satisfy*

$$5 \in F_O, \qquad |F_O| = q - 1, \qquad |F_V| = s - 1.$$

*Then this configuration is impossible.*

*Proof.* String commutation makes $\delta$ centralize $H$. It therefore permutes actual $H$-orbits. Since it preserves $O$ and fixes $D, b$, it also preserves $V$. The actual pure-overlap argument gives

$$x = b^\sigma \ne b, \qquad \delta x = x, \qquad Kx = x, \qquad (6\ b\ x)^{\pm 1} \in \langle \tau, \sigma \rangle. \tag{1}$$

These are the acquired conclusions with natural $R$ and full ambient IP. They require no transitivity of $G_{\ge 4}$. The point $x$ is outside $D, b$ and cannot belong to the free orbit $O$, so lies in $V$. As $\delta|_V$ centralizes a transitive action and fixes $x$, it fixes all of $V$.

The internal pair-forest equality, including the first assigned colour, gives

$$F_O = \{5, \dots, q + 3\}, \qquad I = F_V = \{q + 4, \dots, r - 1\}. \tag{2}$$

On the $q$ pairs of $\delta|_O$, the marks in $F_O$ are the successive transpositions of a path in this order or its reverse. Every other tail mark has trivial pair image. Thus an edge of a colour in $I$ on $O$ can only join a $\delta$-pair; it is already joined by $\tau \in G_j$ and is ambient ineligible. The seven fixed points carry no tail edges. All eligible edges of the colours in $I$ are confined to $V$.

First suppose $s \ge 2$, so $I$ is nonempty, and put $T = G_I$. Since $K$ fixes $x$ but $H$ is transitive on $V$, the mark $\rho_5$ must move $x$ to a distinct point $y \in V$. Every mark in $I$ has index at least $q + 4 \ge 7$, so $T$ commutes with $\rho_5$. It fixes $x$ as a subgroup of $K$ and hence also fixes $y$. Moreover it acts nontrivially on $V$: each colour in $I$ has two assigned eligible edges there. Consequently $o_T(V) \ge 3$, counting the two fixed points and at least one nontrivial orbit. The complete acquired localization theorem gives

$$2s - 2 = 2|I| \le |V| - o_T(V) \le 2s - 3,$$

a contradiction. This argument neither identifies a selected equality forest nor discards its possible square components.

It remains that $s = 1$, so $V = \{x, y\}$ and $I$ is empty. The group $K$ fixes both points of $V$. Removing colour 5 from the pair path in (2) leaves one isolated pair and the natural $S_{q-1}$ path on the remaining pairs. In particular any $K$-fixed point of $O$ must belong to that isolated pair; $q - 1 \ge 2$.

The groups $H$ and $\langle\tau\rangle$ preserve $O$. Since $R = \langle\tau, \sigma, H\rangle$ is transitive on its natural support, $\sigma$ cannot preserve $O$. It fixes $D$ and swaps $b, x$. The only remaining point outside $O$ is $y$, so it must swap $y$ with some $z \in O$. Commutation with $K$ makes $z$ fixed by $K$. The mate $z' = z^\delta$ is also fixed by $K$, and these are exactly the points of the isolated pair. Thus the full $K$-fixed set is

$$D \cup \{b, x, y, z, z'\}.$$

The involution $\sigma$ preserves this set, fixes $D$, and has already paired $b, x$ and $y, z$. It must fix $z'$.

The product $\tau\sigma$ is now a three-cycle on each of the two disjoint invariant sets $\{6, b, x\}$ and $\{y, z, z'\}$. Every odd-order element of the dihedral group $\langle\tau, \sigma\rangle$ belongs to its cyclic rotation subgroup. A power trivial on the second triple is divisible by three and is trivial on the first. This excludes the pure three-cycle asserted in (1), the final contradiction. □

**Attribution and boundary.** The normalized budget, internal pair-path equality, pure head overlap and complete ambient localization are acquired inputs. The new step is the second $T$-fixed anchor supplied by $\rho_5$, followed by the literal two-point residual. The proof retains all square components and uses no projected IP or signed lift classification. Other seven-fixed-point budgets, pairs split between different $H$-orbits, global separator transfer and the complete L044 recovery remain separate. No new alternating rank induction, computation, formal declaration, original or question credit is claimed.